

Math728J – MPI Homework 1

1. Send N real numbers from process 0 to process 1 using blocking communication routines, and measure the elapsed time by the method discussed in **Example 2**. Report the times and the speeds of communication “MBytes/seconds” (the size of a real number is 4-bytes) for $N = 10^1, 10^2, 10^3, 10^4, 10^5, 10^6$.
2. Redo Problem 1 using non-blocking communication routines.
3. In each process, create an array $\mathbf{A}[N]$ with random elements whose values are within $[0, 1]$. We shift $\mathbf{A}$ to the right process and update it using the new $\mathbf{A}$ received from the left process in a group of P processes. Write your program to do this by using two ways:
 - `MPI_SEND` and `MPI_RECV`
 - `MPI_SENDRECV_REPLACE`

Your program should be general for reasonable values of N and P . Print out the $\mathbf{A}$ in each process before and after the shifting for $N = 5$ and $P = 1, 2, 4, 6$.

4. Create a $N \times N$ square matrix with random elements whose values are within $[0, 1]$. Compute the summation of all its entries by using P processes and 1D block partition, and store this value in process 0 and print it out. Measure the elapsed time for computation and communication. Your program should be general for reasonable values of N and P , for example, the values given below. Describe your algorithm and plot the speedup curves for $N = 100, 1000, 2000, 4000$ and $P = 1, 2, 4, 6$ processors.